Supervision 2

Review of MA140 Assignment

- Good overall!
- No one got 'full' clarity marks
  ↳ there's always room for improvement
  (4.5 rounds to 5)
- Just (if possible) make the PDFs as readable as possible
  ↳ scan rather than photograph

Well done!

Many of you overcomplicated Q3. Let's work through the mark scheme:

Q3. $n \in \mathbb{N}$, $x, y > 0$ Prove (by contrapositive)

$$x^n = y^n \Rightarrow x = y$$

Proof. contrapositive: $x \neq y \Rightarrow x^n \neq y^n$

↓

If $x \neq y$, $x < y$ or $y < x$

↓

$0 < x < y \rightarrow 0 < x^2 < y^2$

⋮

$\Rightarrow x^n < y^n$

Why is positivity needed here?

Similarly for $y < x$

$\Rightarrow x^n \neq y^n$.

□

Bijections $f: X \to Y$

Def^n. If $\forall x, x' \in X$

$$f(x) = f(x') \implies x = x'$$

then f is <u>injective</u>

Examples:

			inj.	sur.
	$f: \mathbb{R} \to \mathbb{R}$	$f(x) = x^2$	×	×
	$f: \mathbb{N} \to \mathbb{N}$	$f(x) = x^2$	✓	×
(*)	$f: \mathbb{C} \to \mathbb{C}$	$f(x) = x^2$	×	✓
	$f: \{0,1,2\} \to \{0,1,2\}$	$f(x) = \begin{cases} 1 & \text{if } x=0 \\ 0 & \text{if } x=1 \\ 2 & \text{if } x=2 \end{cases}$	✓	✓

Def^n. $\forall y \in Y \quad \exists x \in X$ s.t. $f(x) = y$

then f is <u>surjective</u>

Examples: same as before.

(*) Proof of surjectivity:

Pick $y \in Y$. $y = re^{i\theta}$ $(r \geq 0)$

define $x = \sqrt{r}e^{i\frac{\theta}{2}}$

then $f(x) = (\sqrt{r}e^{i\frac{\theta}{2}})^2$

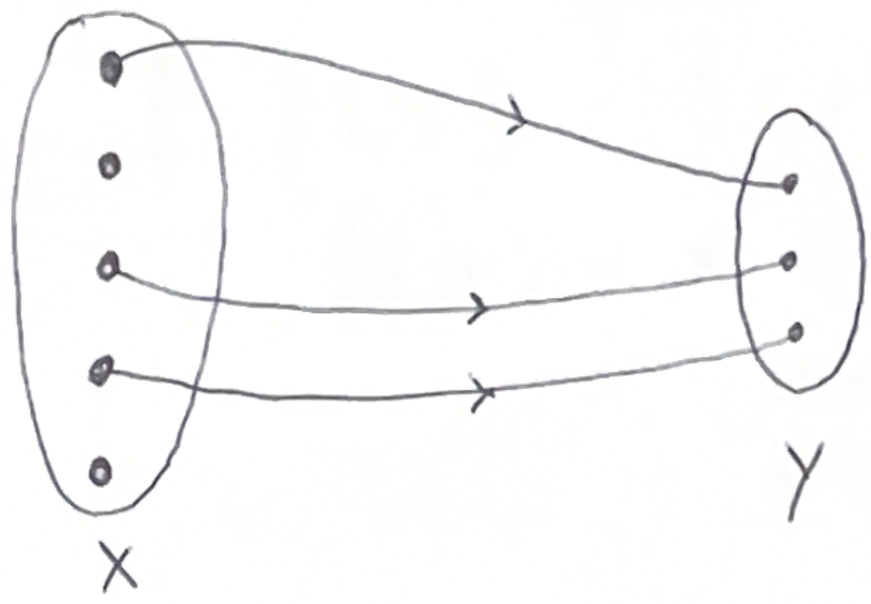
$= re^{i\theta} = y.$

thus $\exists x$ by construction

Def^{n}. f is a <u>bisection</u> if it is injective and surjective.

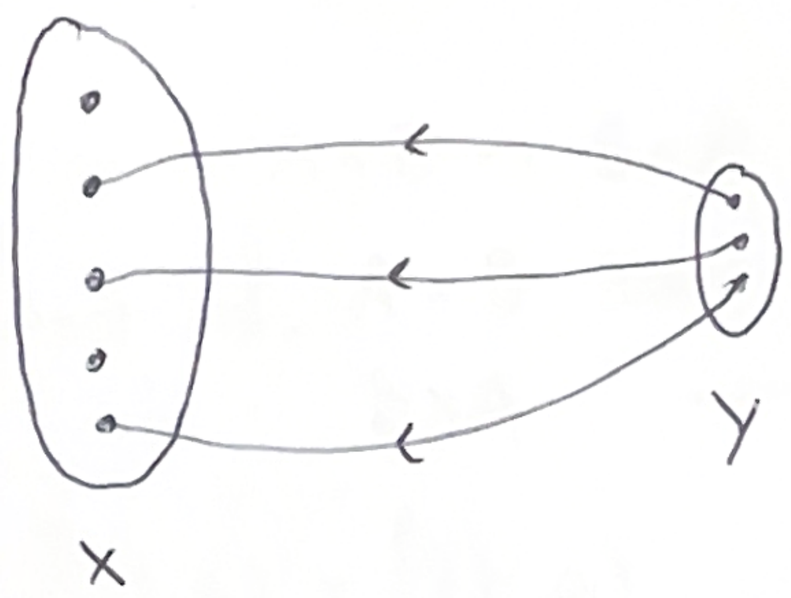
Picture:

Injective:
(1)



} why are these not functions?

Surjective:
(2)



(1) if we tried to make this a function, injectivity fails

(2) indeed, we have the same problem here

$\rightsquigarrow$ pigeonhole principle

for injectivity, we require $|X| \leq |Y|$

by a similar argument (by picture!)

for surjectivity, we require $|X| \geq |Y|$

What would we require for a bisection then? $|X| = |Y|$

Constructing Bijections

Always look for the most simple!

e.g. construct a bij. from $\{1,2\}$ to itself:

$$f: \{1,2\} \to \{1,2\}$$

$$f(x) = x$$

injective ✓
surjective ✓

Examples

ES 2 Q2.3:

Give a bijection $A \times B \to B \times A$

An element of $A \times B$ looks like $(a \in A, b \in B)$

$B \times A$ " ——— " $(b \in B, a \in A)$

$f(a,b) = f(b,a)$ is the simplest.

Injectivity: Sps $f(a,b) = f(c,d)$

$\Rightarrow$ ~~$(a,b) = (c,d)$~~ $(b,a) = (d,c)$

(by properties of ordered pairs) $\Rightarrow$ $a = c$, $b = d$ $\}$ $(c,d) = (a,b)$

Surjectivity: Pick $(b,a) \in B \times A$

Then $b \in B$, $a \in A$ so $(a,b) \in A \times B$

$$f(a,b) = (b,a)$$

ES 2 Q2.6 (exercise for supervisees)

Give a bijection from $(A \times B) \times C \to A \times (B \times C)$

what do the elements look like?

$((a,b),c)$ $(a,(b,c))$

Motivates us to try

$$f((a,b),c) = (a,(b,c))$$

Injectivity: Sps. $f((a,b),c) = f((h,j),k)$

$$\Rightarrow (a,(b,c)) = (h,(j,k))$$

etc.

Surjectivity: as before.

□